

Course outline (Syllabus of the course)

Course Code:CS2500		
Name of the Programme: B.Tech (Hons.)		
Name of the Course: Cloud Computing and Big Data		
Semester: III		
Course credit	No. of hours per week	Total no. of Teaching hours
03	2+0+2	60
Course Objectives:		
Predict suitable cloud computing services and deployment models for different applications.		
Integrate virtualization and containerization technologies to implement CI/CD best practices for deploying cloud-based applications.		
Distinguish the services offered by various Cloud Service Providers across Compute, Storage, Network, and Database categories.		
Analyze the applications of big data analytics, and NoSQL databases to derive insights for decision-making.		
Design solutions for processing big data using the Hadoop Framework in real-world applications.		

Syllabus:

Module - 1 Introduction to Cloud Computing	No. of hours
	4
Introduction to Computing, Characteristics of cloud computing, Types of Cloud Computing Services: PaaS, IaaS, and SaaS, Types of Cloud Deployment Models: Public Clouds, Private Clouds, Community Clouds, community and Hybrid Clouds, Cloud, Grid, Edge, Fog, Utility. Multi-tenancy.	

Module - 2 Virtualization and Containerization	No. of hours
	8
Introduction to virtualization technologies, Benefits of Virtualization, Virtual Machines, Type 1 and Type 2 Hypervisors. Taxonomy of virtualization techniques: machine reference model, Hardware virtualization techniques (full, para, hardware-assisted), OS level virtualization, Programming level virtualization, Application virtualization, storage virtualization, Network virtualization (SDN,NFV), Desktop virtualization, Application server virtualization. Containerization versus virtualization, Introduction to docker, docker hub and Kubernetes. Container orchestration concepts.	

Module - 3 Cloud Services and AWS Platform	No. of hours
	8
Cloud Reference Architecture, IAM (Amazon IAM), Compute services (AWS EC2, AWS Lambda), Storage Services (AWS S3), Database services (Amazon RDS, Dynamo), Content Delivery Services (Amazon CloudFront), Analytics Services (Hadoop Framework and AWS MapReduce), Deployment Services (Amazon Elastic Beanstalk, cloud formation).	

Module - 4 Big Data Analytics and MongoDB	No. of hours
	4
Big Data, Concepts, Properties, characteristics, Applications and value of big data, big data analytics lifecycle, Scalability of Big Data Analytics (Horizontal and Vertical), Designing Data Analytics Architecture, Overview of NoSQL Database, NoSQL to Manage Big Data, Application of MongoDB in Big Data.	

Module - 5 Hadoop Ecosystem	No. of hours
	6
Introduction, Hadoop and its Ecosystem, Hadoop Distributed File System, MapReduce Framework and Programming Model, Hadoop Yarn, Hadoop Ecosystem Tools, HDFS Design Features, HDFS User Commands.	

Lab Programs		[30 Hours]
1	Exploring CloudSim: Understanding Cloud Simulation Through a Basic Setup	
2	Understand CloudSim configurations and scheduling algorithms.	
3	Installing a Type 2 Hypervisor	
4	Create and Manage Virtual Machines	
5	Install Docker and Run Basic Containers	
6	Build and Run Custom Docker Images	
7	Launch an AWS EC2 Instance and Connect via SSH. Understanding AWS IAM Users, Groups, and Roles	
8	Use AWS S3 to Store and Retrieve Data	
9	Integrating AWS Lambda with Amazon DynamoDB	
10	Building a Serverless Chatbot with API Gateway, Lambda, and Amazon Bedrock.	
11	MongoDB Installation and CRUD Operations	
12	Hadoop Setup and WordCount Program Execution	

Course outcomes
CO1: Predict suitable cloud computing services and deployment models for different applications.
CO2: Integrate virtualization and containerization technologies to implement CI/CD best practices for deploying cloud-based applications.
CO3: Distinguish the services offered by various Cloud Service Providers across Compute, Storage, Network, and Database categories.
CO4: Analyze the applications of big data analytics, and NoSQL databases to derive insights for decision-making.

CO5: Design solutions for processing big data using the Hadoop Framework in real-world applications.

Text books	
1.	Rajkumar Buyya, Christian Vecchiola, and Thamarai Selvi - Mastering Cloud Computing - Foundations and Applications Programming (2013), Morgan Kaufmann Publications, ISBN-13: 978-0-12-411454-8, ISBN-10: 0124114547
2.	Toby Velte, Anthony Velte, Robert Elsenpeter, Cloud Computing: A Practical Approach (2017), MC Graw Hill Publications, ISBN-10: 0070683514, ISBN-13: 978-0070683518
3.	Arshdeep Bahga, Vijay Madisetti, Cloud Computing: A Hands-On Approach Hardcover (2014), Vijay Madisetti Publications, ISBN-10: 0996025502, ISBN-13: 978-0996025508
4.	Seema Acharya: Big Data and Analytics (2019), Wiley, ISBN: 8126588365

Reference books	
1.	Rajkumar Buyya, James Broberg, Andrzej Goscinski: Cloud Computing Principles and Paradigms (2013), Edition - 1st, Wiley Publications, ISBN-10: 9788126541256, ISBN-13: 978-8126541256
2.	Brian J.S. Chee, Curtis Franklin Jr., Cloud Computing: Technologies and Strategies of the Ubiquitous Data Center (2010), 1st Edition, CRC Press Publications, 2014 ISBN-10: 1439806128, ISBN-13: 978-1439806128
3.	Balamurugan Balusamy, Nandhini Abirami R, Seifedine Kadry, Amir H. Gandomi: Big Data: Concepts, Technology, and Architecture